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Elastic meta-analytic-predictive prior for dynamically borrowing information from historical data with application to biosimilar clinical trials

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ABSTRACT

A biosimilar is a biological product that is highly similar to and has no clinically meaningful differences from an approved reference product. Focusing on two-arm randomized clinical trials that aim to establish the equivalence between a test biosimilar product and the reference product, we propose the elastic meta-analytic-predictive (EMAP) prior method to leverage rich historical data available on the reference product to improve the power of the biosimilar trials. We first extract the prior information from multiple historical studies through meta-analysis, and then we discount the resulting meta-analytic-predictive (MAP) prior adaptively according to the congruence between the historical reference data and the trial reference arm data via an elastic function. The EMAP prior method is information-borrowing consistent in that asymptotically it achieves full information borrowing when trial reference arm data are congruent to historical reference data, and no information borrowing when trial reference arm data are not congruent to historical reference data. As a result, the method asymptotically controls the type I error rate at the nominal value. Extensive simulation studies show that the EMAP prior outperforms the robust MAP prior. The EMAP prior generates comparable or higher power and provides better-controlled type I errors. We illustrate the proposed methodology using two trial examples.

1. Introduction

According to the United States Food and Drug Administration (FDA) guidance, a biosimilar is a biological product that is highly similar to and has no clinically meaningful differences from an existing FDA-approved biological product (reference product). Other agencies (e.g., EMA and PMDA) have defined biosimilars similarly. Biosimilar products offer additional treatment options for patients and potentially lower health care costs. The FDA recommends a totality of evidence approach based on a stepwise process for the development of biosimilars, starting with analytical characterization, followed by nonclinical characterization, clinical pharmacology (pharmacokinetics (PK)/pharmacodynamics (PD)) and, finally, at least one comparative clinical study in a single representative indication to confirm similar efficacy, safety and immunogenicity with the reference product. In the comparative clinical efficacy and safety study, patients are randomized to either a reference product arm or a biosimilar arm, and the objective is to show that the efficacy of the biosimilar is equivalent to the reference product. At the time of designing biosimilar clinical trials, the reference product has been approved by regulatory agencies and on market for years. It is of great interest to leverage these abundant historical data to enhance the efficiency of the proposed biosimilar clinical trial.

Various methods have been developed to borrow information from historical data to improve the efficiency of clinical trials. Ibrahim et al. [1] proposed the power priors method to incorporate historical data from previous studies. Hobbs et al. [2] proposed the commensurate prior to incorporate historical data in clinical trials. Schmidli et al. [3] proposed the Bayesian meta-analytic-predictive (MAP) prior to borrow information from multiple historical datasets and robust MAP priors through mixtures of distributions. Chen et al. [4] proposed Bayesian methods, including hierarchical priors and power priors, to incorporate historical data in the design of non-inferiority clinical trials. Jiang, Nie and Yuan [5] proposed the elastic prior to adaptively borrow information from historical data based on their congruence with current trial

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data, showing better operating characteristics than power prior and commensurate prior. In addition, several methods have been developed specifically for biosimilar trials. Pan, Yuan, and Xia [6] proposed a calibrated power prior approach to adaptively borrow information from historical study for the two-arm randomized biosimilar trials. The calibrated power prior requires subjective-level historical data, which often are not available in biosimilar studies. Weiss et al. [7] proposed to use a MAP prior for the reference product, and proposed priors for the biosimilar informed by the strength of the similarity evidence between the proposed biosimilar product and the reference product generated in the earlier steps of the biosimilar development group from analytical and clinical PK/PD studies. The fundamental challenge for information-borrowing methods is about how to efficiently borrow information when the historical reference data and the reference arm data in the current trial are congruent or comparable, while minimizing the bias and inflation of type I errors when the two data are conflicted or incongruent.

We propose an elastic meta-analytic-predictive (EMAP) prior to dynamically borrow information from historical reference data. The EMAP induces strong information borrowing when the historical reference data are congruent to the trial reference arm data, and prevents information borrowing when the historical reference data are conflicted with the trial reference arm data. The EMAP achieves such dynamic information borrowing by adaptively discounting the prior information using an elastic function, which is defined as a monotonic function of congruence measure between historical data and trial data. Our simulation studies show that, compared to some existing methods (e.g., robust MAP), EMAP has better operating characteristics. It improves the power when the historical data are congruent to the trial data, and has better type I error control when the two data sources are incongruent. Our method is an extension of the elastic prior methodology [5] to biosimilar studies.

Compared to the calibrated power prior proposed by Pan, Yuan, and Xia [6], the EMAP is more general in that it (1) accommodates multiple historical datasets, (2) handles both the case that patient-level data are available and the case that only study-level summary statistics are available, and (3) allows parameter-specific adaptive information borrowing (the calibrated power prior does not allow this as it discounts the whole likelihood function). Compared to the method of Weiss et al. [7], which always fully borrows the historical information, the EMAP is adaptive and borrows more (or less) information from the historical data when the reference arm is more (or less) similar to the historical data.

This article proceeds as follows. Section 2 introduces Bayesian equivalence and noninferiority tests, proposes the EMAP prior for binary and continuous outcomes, and studies its statistical properties. Section 3 applies the proposed methodology to two trial examples and conducts simulation studies to investigate the operating characteristics of the proposed EMAP design. We close with a brief discussion in Section 4.

2. Method

2.1. Bayesian equivalence and noninferiority test

Consider a two-arm biosimilar trial where patients are randomized to a reference product arm and a test (biosimilar) product arm. Let y denote the efficacy endpoint that is a binary variable following a Bernoulli distribution or a continuous variable following a normal distribution. Let μ_r and μ_t denote $E(y)$ for the reference and test arms, respectively. To establish that the test product is equivalent to the reference product, the frequentist approach is, for a binary endpoint, to perform the hypothesis testing H_0: $\mu_t/\mu_r \le \lambda_L$ or $\mu_t/\mu_r \ge \lambda_U$ versus H_1: $\lambda_L < \mu_t/\mu_r < \lambda_U$; and for a continuous endpoint that follows a normal distribution, to perform the hypothesis testing H_0: $\mu_t - \mu_r \le \lambda_L$ or $\mu_t - \mu_r \ge \lambda_U$ versus H_1: $\lambda_L < \mu_t - \mu_r < \lambda_U$, where λ_L and λ_U are the prespecified margins chosen based on the endpoint, product and disease.

Under the Bayesian paradigm, equivalence can be evaluated based on the following posterior probability criterion [6]: the test product and the reference product are deemed equivalent if $\Pr(\lambda_L < \mu_t/\mu_r < \lambda_U | data) > C$ or $\Pr(\lambda_L < \mu_t - \mu_r < \lambda_U | data) > C$, depending on the definition of equivalence, where C is a probability cutoff calibrated through simulations to ensure a desirable type I error. Asymptotically, setting $C = 1 - \alpha$ leads to a test with a type I error rate of α, providing the connection between Bayesian and frequentist approaches. This Bayesian criterion can also be used to determine whether the test product is noninferior to the reference product by setting $\lambda_U = \infty$, assuming that a higher value of μ_t indicates a better treatment effect.

2.2. Meta-analytic-predictive prior

Consider a binary endpoint y, and let $\boldsymbol{D}_0 = (D_{01}, \cdots, D_{0K})$ denote K independent historical data for the reference product, where $D_{0k} = (x_k, n_k)$ with x_k and n_k denoting the number of responses and sample size of the kth historical dataset, respectively. In what follows, we focus on how to leverage $\boldsymbol{D}_0$ to improve the posterior inference for the reference arm. The posterior inference for the test arm can be made based on the Bayes' theorem with a noninformative prior.

One way to aggregate the prior information of $\boldsymbol{D}_0$ is to use meta-analysis based on a Bayesian hierarchical (or random-effect) model [7],:

$$x_k|\mu_k \sim \text{Binomial}(n_k, \mu_k), \tag{1}$$

$$\mu_k|\alpha_0, \beta_0 \sim \text{Beta}(\alpha_0, \beta_0),$$

where μ_k is the response rate for D_{0k}, $k = 1, \cdots, K$, and α_0, $\beta_0 > 0$ are unknown parameters. To facilitate the specification of priors for α_0 and β_0, we reparameterize them using the mean (μ_0) and variance (τ_0^2) parameters,

$$\mu_0 = \frac{\alpha_0}{\alpha_0 + \beta_0}$$

$$\tau_0^2 = \frac{\mu_0(1-\mu_0)}{\alpha_0 + \beta_0 + 1},$$

and then assign μ_0 and τ_0^2 appropriately vague prior distributions to cover the range of plausible values of these parameters, which depend on the drug and patient population under investigation. For example, Weiss et al. [7] used the priors $\mu_0 \sim Beta(12, 12)$ and $\tau_0 \sim Beta(0.8, 7.2)$.

By treating the current reference arm as the $(K+1)$th study of the meta-analysis and fitting the above Bayesian hierarchical model, we obtain the MAP prior for the reference arm, given by $p(\mu_r|\boldsymbol{D}_0) \equiv p(\mu_{(K+1)}|\boldsymbol{D}_0)$. This posterior predictive distribution does not have a closed form, but can be approximated using a beta distribution

$$p(\mu_r|\boldsymbol{D}_0) = \text{Beta}(\alpha, \beta), \tag{2}$$

where α and β are determined by matching the first two moments of $Beta(\alpha, \beta)$ with those of $p(\mu_{(K+1)}|\boldsymbol{D}_0)$, that is,

$$\frac{\alpha}{\alpha+\beta} = M_1 \tag{3}$$

$$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = M_2$$

where M_1 and M_2 are the mean and variance of $p(\mu_{(K+1)}|\boldsymbol{D}_0)$. Weiss et. al [7] showed that this approximation is sufficiently accurate in application, thus we use it as the MAP prior hereafter. When necessary, a mixture of beta distributions can be used to improve the accuracy of approximation [3].

In the special case that there is only a single historical data $\boldsymbol{D}_0 \equiv D_{01}$, rather than conducting meta-analysis, the standard beta-binomial model can be employed to model D_{01} and use the resulting posterior as the prior of μ_r:

$$p(\mu_r|D_{01}) = \text{Beta}(\alpha, \beta), \tag{4}$$

where $\alpha = \alpha_0 + x_1$ and $\beta = \beta_0 + n_1 - x_1$, with α_0 and β_0 being small constants (e.g, 0.01).

To incorporate historical data $\boldsymbol{D}_0$, we multiply the MAP prior with the likelihood of the reference arm data $D = (x_r, n_r)$, and the posterior distribution of the response rate μ_r for the reference arm arises as

$$\mu_r|D, \boldsymbol{D}_0 \sim \text{Beta}(\alpha + x_r, \beta + n_r - x_r).$$

Such full information borrowing is sound when D is congruent to $\boldsymbol{D}_0$, but inappropriate when they are incongruent, leading to inflated type I errors or lower powers, depending on the direction that D conflicts with $\boldsymbol{D}_0$.

2.3. Elastic MAP prior

To address this issue, we propose the elastic meta-analytic-predictive (EMAP) prior to dynamically borrow information from historical data $\boldsymbol{D_0}$ according to the degree of congruence between D and $\boldsymbol{D_0}$. When D and $\boldsymbol{D_0}$ are congruent, EMAP induces strong information borrowing; and when D and $\boldsymbol{D_0}$ are incongruent, EMAP prevents information borrowing.

We first provide two key definitions in this method. We define that D and $\boldsymbol{D_0}$ are congruent when they have the same distribution, i.e., $\mu_r = \mu_0$ in the settings described earlier, and incongruent when $\mu_r \neq \mu_0$. We define a measure of congruence S as a statistic of D and $\boldsymbol{D_0}$ that quantifies the strength of evidence for the congruence between D and $\boldsymbol{D_0}$. S is defined such that it asymptotically converges to 0 when $\mu_r = \mu_0$, and to ∞ when $\mu_r \neq \mu_0$. For a binary endpoint, given the trial data $D = (y_1, \cdots, y_n)$ and denote the mean of D as $\overline{y_r} = \sum y_i/n$, we propose to measure this strength of congruence by the scaled binomial test statistic:

$$S = n^{-\frac{1}{4}} \frac{|\overline{y_r} - \mu_0|}{\sqrt{\frac{\mu_0(1-\mu_0)}{n}}},$$

where μ_0 is often replaced by its posterior estimate from the meta-analysis. In this S statistic, $n^{-\frac{1}{4}}$ is a scale factor added for the property of information-borrowing consistency, described in Theorem 1 later. The choice of S is not unique, other statistics (e.g., Chi-square statistic) could also be used. Our method is not limited to a specific choice of S.

2.3.1. EMAP prior

With S at hand, the EMAP prior is defined as

$$p(\mu_r|\boldsymbol{D}_0) \sim \text{Beta}(\alpha g(S), \beta g(S))$$

where α and β are given by Equation (3), and $g(S) \in [0, 1]$ is an elastic function that serves as a variance inflation factor (or information discount factor) to control the information borrowing, according to the value of S. If $g(S) = 1$, EMAP becomes MAP and executes full information borrowing, and if $g(S) = 0$, no information will be borrowed from $\boldsymbol{D_0}$. The role of the elastic function is to ensure that the degree of information borrowing elastically changes with the level of congruence.

Any monotonic function that maps S to [0, 1] can be used as an elastic function, as long as it satisfies the condition that $g(S) \to 1$ when D and $\boldsymbol{D_0}$ are congruent, and $g(S) \to 0$ when D and $\boldsymbol{D_0}$ are not congruent. The elastic function and dynamic information borrowing is illustrated in Fig. 1. Following Jiang et al. [5], one elastic function that yields good operating characteristics is the logistic function:

$$g(S) = \frac{1}{1 + exp\{a + b \times log(S)\}}, \tag{5}$$

where a and $b > 0$ are tuning parameters that can be calibrated using the following steps.

1. (*Congruence case*) Generate 10,000 replicates of the reference arm data $D = (x_r, n_r)$ from $x_r \sim Binomial(n_r, \mu_0)$, and calculate statistic S for each replicate. Let S_{ρ_1} denote the ρ_1 percentile of the empirical distribution of S.
2. (*Incongruence case*) (2a) Specify lower and upper margins γ^- and γ^+ (e.g., $\gamma^- = 0.75$ and $\gamma^+ = 1.25$) such that when $\mu_r/\mu_0 \leq \gamma^-$ and $\mu_r/\mu_0 \geq \gamma^+$, D and $\boldsymbol{D_0}$ should be regarded as clinically significantly

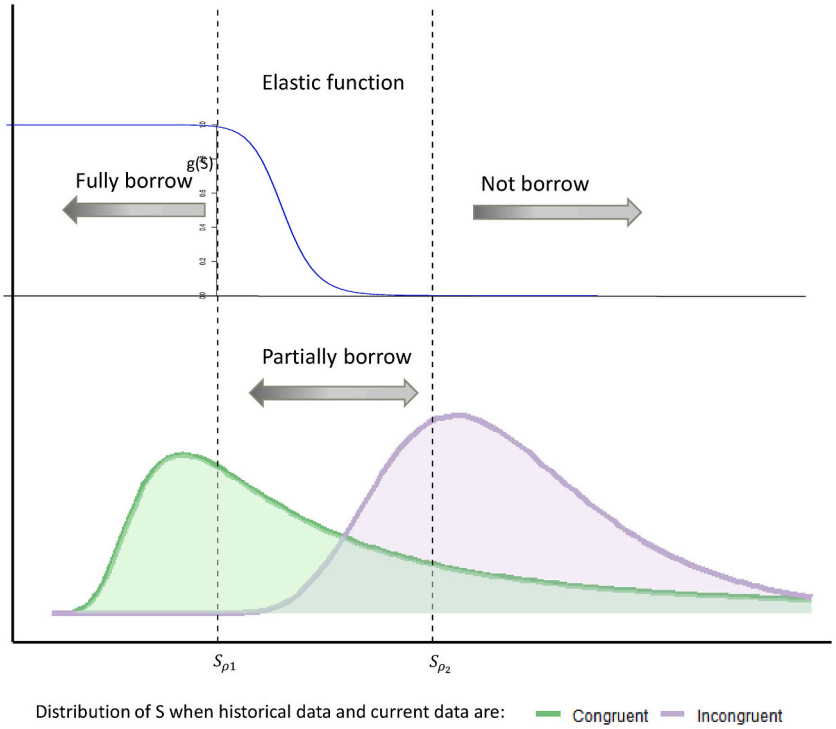


Fig. 1. The mechanism of EMAP prior to achieve dynamic information borrowing through the elastic function. Two curves at the bottom show the distribution of the congruence measure when historical data are congruent and incongruent, respectively, to simulated trial data.

different, and thus little information should be borrowed. (2b) Generate 10,000 replicates of the reference arm data $D = (x_r, n_r)$ from $x_r \sim Binomial(n_r, \gamma^- \mu_0)$, and calculate statistic S for each replicate, denoted as S^-. Let $S^-_{\rho_2}$ denote the ρ_2 percentile of the empirical distribution of S^-. Repeat this step by replacing γ^- with γ^+, and let $S^+_{\rho_2}$ denote the ρ_2 percentile of the empirical distribution of S^+. Define $\widetilde{S}_{\rho_2} = \min(S^-_{\rho_2}, S^+_{\rho_2})$.

3. Solve a and b based on the following two equations:

$$H_1 = g(S_{\rho_1}) \tag{6}$$

$$H_2 = g(\widetilde{S}_{\rho_2}), \tag{7}$$

where H_1 is a constant close to 1 (e.g., 0.99) and H_2 is a constant close to 0 (e.g., 0.01). Equation (6) ensures strong information borrowing in the congruent case and Equation (7) prevents information borrowing in incongruent case. This leads to the following solution:

$$a = log(\frac{1-H_1}{H_1}) - \frac{log(\frac{(1-H_1)H_2}{(1-H_2)H_1}) log(S_{\rho_1})}{log(\frac{S_{\rho_1}}{\widetilde{S}_{\rho_2}})}$$

$$b = log(\frac{(1-H_1)H_2}{(1-H_2)H_1}) \Big/ log(\frac{S_{\rho_1}}{\widetilde{S}_{\rho_2}}).$$

In Step 2, we may set γ^- and γ^+ at equivalence margins, i.e., $\gamma^- = \lambda_L$ and $\gamma^+ = \lambda_U$, but not mandatory. For example, if it is desirable to borrow information as long as $\lambda_L < \mu_r/\mu_0 < \lambda_U$ (e.g., we intend to achieve some information borrowing when μ_r/μ_0 is close to λ_L and λ_U), then we could set γ^- and γ^+ slightly away from λ_L and λ_U, e.g., $\gamma^- = 0.5\lambda_L$ and $\gamma^+ = 1.5\lambda_U$

In Steps 1 and 2, ρ_1 and ρ_2 define the borrowing and no borrowing regions (see Fig. 1). We may simply choose $\rho_1 = \rho_2 = 0.5$, i.e., median of the empirical distribution of S. A better and more flexible approach is to choose ρ_1 and ρ_2 to maximize the trade-off between the power (in the congruent case) and type I error (in the incongruent case). Specifically, let ψ denote the power under the congruent case, and let φ denote the type I error under the incongruent case described in Step 2. We define a power and type I error trade-off utility function:

$$U(\rho_1, \rho_2) = \psi - w_1\varphi - w_2 * (\varphi - r) * I(\varphi > r) \tag{8}$$

where w_1 and w_2 are pre-specified penalties. This utility imposes a penalty of w_1 for each unit increase of a type I error before it reaches r, and a penalty of $w_1 + w_2$ after it reaches r. In our simulation and application, we set $w_1 = 1$, $w_2 = 2$, and $r = 0.1$, which means that before the type I error reaches 0.1, the penalty for a 1% increase of type I errors is to deduct the power by 1%; and once the type I error exceeds 0.1, the penalty for a 1% increase of type I errors grows to deduct the power by 3%. If it is desirable to control the type I error rate at r under the incongruence case ($\mu_r/\mu_0 \le \gamma^-$ and $\mu_r/\mu_0 \ge \gamma^+$), we can simply set w at a very large value (e.g., 100). The choice of utility is not unique and should reflect the trial objectives. For example, if the primary interest is estimation, we may use the mean square error as $U(\rho_1, \rho_2)$. The values of ρ_1 and ρ_2 that maximize $U(\rho_1, \rho_2)$ can be determined using a grid search, and details are provided in the Appendix.

A special and limiting form of the elastic function Equation (5), obtained when b is large, is the following step function:

$$g(S) = \begin{cases} 1 & S \le S_{\rho_1} \\ 0 & S > S_{\rho_1} \end{cases} \tag{9}$$

where we fully borrow information from the meta-analytic-predictive prior if $S \le S_{\rho_1}$ and don't borrow information if $S > S_{\rho_1}$ (see Fig. 2b). Numerical studies show that this step function could achieve comparable operating characteristics as the smooth elastic function. The advantage of the step elastic function is that it is simpler to calibrate using two steps,

1. (*Congruence case*) Generate 10,000 replicates of the reference arm data $D = (x_r, n_r)$ from $x_r \sim Binomial(n_r, \mu_0)$, and calculate statistic S for each replicate. Let S_{ρ_1} denote the ρ_1 percentile of the empirical distribution of S.
2. Conduct a grid search to identify the ρ_1 that optimizes the utility function in Equation (8).

Because of its simplicity, the step elastic function may be preferred in practice.

2.3.2. Properties of EMAP prior

The proposed EMAP has several desirable properties.

Theorem 1. *Given that S is consistent for testing the congruence of* $\boldsymbol{D_0}$ *and*

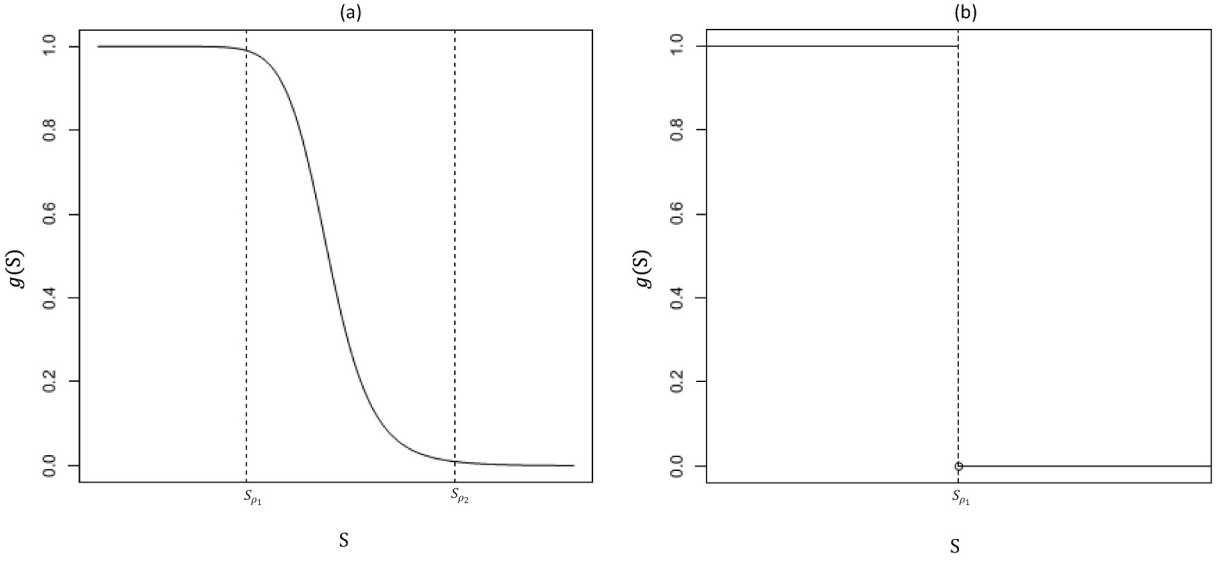


Fig. 2. (a) Smooth elastic function given by Equation (5), and (b) step elastic function given by Equation (9).

D. EMAP prior is information-borrowing consistent in that when the sample size of current reference arm $n_r \to \infty$, it achieves full information borrowing if $\boldsymbol{D_0}$ and D are congruent (i.e., $\mu_0 = \mu_r$) and does not borrow any information if $\boldsymbol{D_0}$ and D are incongruent (i.e., $\mu_0 \neq \mu_r$).

Corollary 1. *The EMAP prior asymptotically maintains the type I error at the nominal value.* The proof of Theorem 1 and Corollary 1 are provided in Appendix. The biggest concern and barrier for adopting information borrowing methods is the potential bias and risk of inflating type I errors caused by the information borrowing when $\boldsymbol{D_0}$ and D are incongruent. Theorem 1 shows that the EMAP can asymptotically maintain type I errors at the nominal value when $\boldsymbol{D_0}$ and D are not congruent. In contrast, most existing dynamic information borrowing methods, including the power prior, commensurate prior, and robust MAP prior do not have this property. To achieve the information-borrowing consistency, they typically require that the number of historical datasets (i.e., K) goes to infinity, which is usually unrealistic.

In addition, it is straightforward to calculate the effective sample size of the EMAP prior, which is simply $m \times g(S)$, where m is the effective sample size of the MAP prior. If the approximation of beta distribution Equation (3) is used, $m = \alpha + \beta$ [11,12]. Lastly, the specification of required parameters that define the EMAP prior does not depend on the trial data, since all data are simulated based on the meta-analysis on $\boldsymbol{D_0}$. Thus, the EMAP prior satisfies an essential requirement for adaptive trials that "potential adaptive design modifications should be planned and described in the clinical trial protocol (and in a separate statistical analysis plan) prior to initiation of the trial," as specified in the FDA's Guidance for Industry on Adaptive Designs for Clinical Trials of Drugs and Biologics.

2.3.3. Continuous endpoint

The proposed methodology is also applicable to a continuous endpoint. Suppose for the kth historical dataset, summary statistics $D_{0k} = (x_k, s_k, n_k)$ are available, where x_k and s_k denote the sample mean of the endpoint and its standard deviation, and n_k denotes the sample size of each study, $k = 1, \cdots, K$. We first perform meta-analysis based on a Bayesian normal-normal hierarchical model:

$$x_k|\mu_k, \sigma_k^2 \sim N(\mu_k, \sigma_k^2) \tag{10}$$

$$\mu_k|\mu_0, \tau_0^2 \sim N(\mu_0, \tau_0^2),$$

where μ_k and σ_k^2 are mean and variance parameters for D_{0k}, and μ_0 and τ_0 are unknown model parameters. Replacing σ_k^2 with s_k^2/n_k and assigning noninformative priors $\mu_0 \sim N(0, 10^5)$ and $\tau_0^{-2} \sim \Gamma(0.001, 0.001)$, the MAP is obtained as

$$p(\mu_r|\boldsymbol{D}_0) \equiv p(\mu_{(K+1)}|\boldsymbol{D}_0).$$

In the case that patient-level data are available for D_{0k}, a similar Bayesian hierarchical model can be used for meta-analysis, but replacing x_k by patient-level observation y_{ki} and treating σ_k^2 as an unknown parameter.

This MAP prior does not have a closed form, but can be approximated using a normal distribution

$$p(\mu_r|\boldsymbol{D}_0) = N(\theta, \tau^2),$$

where θ and τ^2 are determined by matching the first two moment of $p(\mu_{(K+1)}|\boldsymbol{D_0})$. The EMAP prior is defined as

$$p(\mu_r|\boldsymbol{D}_0) \sim N(\theta, \tau^2/g(S)),$$

where $g(S)$ is defined previously in Equation (5) or (9). Let s denote the sample standard deviation and n denote the sample size of the reference arm. The congruence measure S is defined as the scaled t test statistic that evaluates whether the mean of reference arm $\overline{y_r}$ is different from its expected mean μ_0 (if it conforms to the meta-analysis model):

$$S = n^{-\frac{1}{4}} \frac{|\overline{y_r} - \mu_0|}{s\sqrt{\frac{1}{n}}}$$

where μ_0 is often replaced by its estimate from the meta-analysis model Equation (10). Along the same line as Theorem 1 and Corollary 1, it can be shown that the resulting elastic prior is information-borrowing consistent and asymptotically maintains the type I error at the nominal value.

Assigning μ_r the EMAP prior and σ^2 the non-informative prior $p(\sigma^2) \propto \sigma^{-2}$, given the trial data $D = (y_1, \cdots, y_n)$, the joint posterior of μ_r and σ^2 is given by

$$p(\mu_r, \sigma^2|D, \boldsymbol{D}_0) \propto (\sigma^2)^{-\frac{n}{2}-1} exp\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu_r)^2\} exp\{-\frac{g(S)}{2\tau^2}(\mu_r - \theta)^2\}.$$

3. Simulation studies

3.1. Two trial examples

3.1.1. Example 1 (Binary endpoint)

Consider a two-arm randomized clinical trial testing whether a test product is biosimilar to the reference product trastuzumab, a monoclonal antibody used to treat patients with breast or stomach cancer who are HER2 receptor positive. The outcome of interest is binary, indicating whether a patient has pathologic complete response after treatment. Six historical datasets are available for trastuzumab; see Table 1 [15,18]. The response rate in these historical datasets varies from 0.38 to 0.58 with the sample size ranging from 42 to 307. We performed a meta-analysis to synthesize the historical data using the random-effect model given by Equation (1), resulting in the MAP prior $\mu_r \sim Beta(122.02, 173.59)$, with a prior mean of $\mu_0 = 0.413$ and a standard deviation of $\tau_0 = 0.029$. The equivalence margin is defined as $0.8 < \mu_t/\mu_r < 1.25$, with margins $\lambda_L = 0.8$ and $\lambda_U = 1.25$.

3.1.2. Example 2 (Continuous endpoint)

Consider a two-arm randomized trial to evaluate whether a test product is biosimilar to the reference product omalizumab, a monoclonal antibody that has been used to control severe allergic asthma or treat chronic spontaneous urticaria. The outcome of interest is the reduction of the weekly itch score, which is considered as continuous and normally distributed. Table 2 reports historical data from seven studies on omalizumab in doses of 300 mg every 4 weeks for treating chronic spontaneous urticaria. Across these studies, the mean of reduction ranges from 8.3 to 11.5 and standard deviation ranges from 4.54 to 11.05, with sample sizes varying from 8 to 252 [14]. We conducted meta-analysis based on the Bayesian normal-normal hierarchical model Equation (10). The resulting MAP prior is $\mu_r \sim N(9.93, 0.62)$. The equivalence is defined as $-2.5 < \mu_t - \mu_r < 2.5$, with margins $\lambda_L = -2.5$ and $\lambda_U = 2.5$.

We evaluate the operating characteristics of the proposed methodology using the two trial examples. In example 1, we simulated the reference arm data $x_r \sim Binomial(n_r, \mu_r)$ and the test arm data $x_t \sim Binomial(n_t, \mu_t)$. We considered $\mu_r = 0.413$, representing that reference

Table 1
Historical data from six clinical trials of trastuzumab with pathologic complete response (pCR) as the primary endpoint.

Study/Author	Year	Number of pCR	Sample size
ABCSG-24	2009	17	42
CALGB	2013	48	120
NOAH	2010	45	117
GeparQuinto	2012	137	307
HannaH	2012	121	298
Bayraktar et al	2012	137	235

Table 2
Mean and standard deviation (SD) of the reduction of weekly itch score in seven clinical trials of omalizumab.

Study/Author	Year	Mean	SD	Sample size
ASTERIA I	2013	9.4	5.73	81
ASTERIA II	2015	9.8	6	79
GLACIAL	2013	10.5	11.05	252
MOA	2015	11.5	4.54	17
MYSTIQUE	2011	9.2	5.98	25
X-ACT	2015	8.3	7.58	44
XCUISITE	2011	11.19	6.46	8

data D are congruent to $\boldsymbol{D_0}$ (i.e., $\mu_r = \mu_0$), and $\mu_r = 0.310$ and 0.516, representing D are incongruence to $\boldsymbol{D_0}$ (i.e., $\mu_r = 0.75\mu_0$ or $\mu_r = 1.25\mu_0$). We considered five response rates for the test arm: $\mu_t = \mu_r$, $0.95\mu_r$, $1.05\mu_r$ $0.8\mu_r$, and $1.25\mu_r$, to represent exactly biosimilar, approximately biosimilar ($\mu_t = 0.95\mu_r$ and $1.05\mu_r$), and not biosimilar ($\mu_t = 0.8\mu_r$ and $1.25\mu_r$) to the reference arm. The total sample size is 840, randomized in a 1:2 ratio to the reference and test arms, that is, $n_r = 280$ and $n_t = 560$. As the power of the test depends on μ_r, under the scenarios with $\mu_r = 0.310$, we used a larger sample size $n_r = 420$ and $n_t = 840$ to ensure that the power is in the range of practical interest (e.g., >50%).

In example 2, we simulated the reference arm data as $y_i \sim N(\mu_r, \sigma_r^2)$, $i = 1, \ldots, n_r$ and the test arm data as $y_i \sim N(\mu_t, \sigma_t^2)$, $i = 1, \ldots, n_t$. We considered $\mu_r = 9.93$, representing that reference data D are congruent to $\boldsymbol{D_0}$ (i.e., $\mu_r = \mu_0$), and $\mu_r = 7.43$ and 12.43, representing D are incongruent to $\boldsymbol{D_0}$ (i.e., $\mu_r = \mu_0 \pm 2.5$). We varied the value of μ_t to simulate that the test product and the reference product are exactly biosimilar with $\mu_t = \mu_r$, approximately biosimilar with $\mu_t = \mu_r \pm 0.5$, and not biosimilar with $\mu_t = \mu_r \pm 2.5$. Under each scenario, the values of σ_r and σ_t are set the same as the pooled standard deviation from seven historical datasets (i.e., $\sigma_r = \sigma_t = \widehat{\sigma_0} = 8.9569$). The total sample size is 375, randomized to the reference and test arm in a 1:2 ratio, i.e., $n_r = 125$ and $n_t = 250$.

We compared the proposed EMAP prior method to the no-borrowing (NB) approach that ignores the historical data, the MAP approach that directly uses the MAP prior for inference, and the robust MAP approach [3] with weights of 0.9 and 0.5 in the mixture prior (denoted as rMAP90 and rMAP50, respectively). For the proposed EMAP prior method, we considered using the smooth elastic function Equation (5) and the step elastic function Equation (9), denoted as EMAP1 and EMAP2, respectively. Under each scenario, we simulated 5,000 trials.

3.2. Simulation results

Table 3 presents simulation results for trial example 1 with a binary endpoint. In scenarios 1–5, $\boldsymbol{D_0}$ and D are congruent. Specifically, in scenarios 1 and 2, μ_t is not biosimilar to μ_r and all methods control type I errors at the nominal value of 5%. In scenarios 3–5, μ_t and μ_r are biosimilar. EMAP1 and EMAP2 generate substantially higher power than NB, and higher or comparable power to rMAP50. For example, in scenario 3, the powers of EMAP1 and EMAP2 are 90.1% and 90.7%, which are 29.0% higher than the NB design. The power of rMAP90 and MAP is slightly higher than EMAP1 and EMAP2, however, it leads to a substantially higher type I error or lower power when D is not congruent to $\boldsymbol{D_0}$ (e.g., scenarios 6–11). Scenarios 6–11 consider the cases that D is not congruent to $\boldsymbol{D_0}$. Specifically, in scenarios 6 and 7, μ_t and μ_r are not biosimilar. EMAP1 and EMAP2 control type I errors substantially better than rMAP50, rMAP90, and MAP. For example, in scenario 6, type I errors of EMAP1 and EMAP2 are 7.2% and 5.8%, respectively, whereas the type I error of rMAP50 doubles that (14 %), and the type I errors of rMAP90 and MAP are 30.3% and 68.9%, respectively. In scenario 7, type I errors of EMAP1 and EMAP2 are 9.2% and 9%, whereas rMAP50, rMAP90, and MAP inflate type I errors to 16.1%, 32.1%, and 55%. In scenarios 8–11, μ_t and μ_r are biosimilar. EMAP1 and EMAP2 yield higher power than rMAP50, rMAP90 and MAP. For example, in scenario 8, EMAP1 and EMAP2 generate powers of 80.0% and 79.4%. In contrast, the powers of rMAP50, rMAP90, and MAP are 60.4%, 42.5%, and 32.7%, respectively. Similar results are observed in scenarios 9–11.

Table 4 displays simulation results for trial example 2 with a continuous endpoint. In general, the results are consistent with those for the binary endpoint. Scenarios 1–5 consider the cases that $\boldsymbol{D_0}$ and D are congruent. In scenarios 1 and 2, μ_t and μ_r are not biosimilar. All methods control type I errors at the nominal value. In scenarios 3–5, μ_t and μ_r are biosimilar. EMAP1 and EMAP2 designs generate substantially higher powers than NB, and higher or comparable powers to rMAP50. For example, in scenario 3, the power of EMAP1 is 90.1%, which is 28.7% higher than NB. Again, rMAP90 and MAP yield slightly higher powers than EMAP1 and EMAP2, but lead to a substantially higher type I error or lower power when D is not congruent to $\boldsymbol{D_0}$ (e.g., scenarios 6–11). In scenarios 6–11, D is not congruent to $\boldsymbol{D_0}$. In scenarios 6 and 7, μ_t and μ_r are not biosimilar. EMAP1 and EMAP2 offer better type I error control than rMAP50, rMAP90, and MAP. For example, in scenario 6, type I errors of EMAP1 and EMAP2 are 10.0% and 9.7%, respectively, whereas the type I errors of rMAP50, rMAP90, and MAP are 19.2%, 39.4%, and 81.3%, respectively. In scenarios 8–11, μ_t and μ_r are biosimilar. EMAP1 and EMAP2 yield higher powers than rMAP50, rMAP90, and MAP designs. For example, in scenario 8, EMAP1 and EMAP2 have powers of 61.9% and 58.8%, respectively, whereas the powers of rMAP50, rMAP90, and MAP are lowered to 42.6%, 25.5%, and 13.6%.

3.3. Noninferiority test

We briefly evaluated the performance of the proposed EMAP method for noninferiority trials based on the historical data of trastuzumab (trial

Table 3
Simulation results of equivalence test for trastuzumab trial with a binary endpoint.

			Probability of claiming biosimilarity					
Scenario	μ_r	μ_t	NB	EMAP1	EMAP2	rMAP50	rMAP90	MAP
			Congruence					
1	0.413	$0.8\mu_r$*	0.051	0.050	0.049	0.051	0.051	0.052
2	0.413	$1.25\mu_r$*	0.049	0.043	0.043	0.045	0.038	0.043
3	0.413	μ_r	0.620	0.901	0.907	0.906	0.941	0.942
4	0.413	$0.95\mu_r$	0.538	0.793	0.792	0.812	0.830	0.838
5	0.413	$1.05\mu_r$	0.572	0.843	0.845	0.86	0.878	0.885
			Incongruence					
6	0.310	$1.25\mu_r$*	0.049	0.072	0.058	0.140	0.303	0.689
7	0.516	$0.8\mu_r$*	0.045	0.092	0.090	0.161	0.321	0.550
8	0.516	$1.05\mu_r$	0.769	0.800	0.794	0.604	0.425	0.327
9	0.516	μ_r	0.864	0.859	0.870	0.791	0.714	0.683
10	0.310	$0.95\mu_r$	0.511	0.575	0.542	0.455	0.251	0.167
11	0.310	μ_r	0.605	0.648	0.633	0.573	0.408	0.433

* Type I error.

Table 4
Simulation results of equivalence test for omalizumab trial with a continuous endpoint.

			Probability of claiming biosimilarity					
Scenario	μ_r	μ_t	NB	EMAP1	EMAP2	rMAP50	rMAP90	MAP
			Congruence					
1	9.930	$\mu_r - 2.5$*	0.049	0.050	0.050	0.049	0.051	0.051
2	9.930	$\mu_r + 2.5$*	0.050	0.051	0.053	0.050	0.051	0.050
3	9.930	μ_r	0.614	0.901	0.892	0.927	0.965	0.977
4	9.930	$\mu_r - 0.5$	0.571	0.851	0.835	0.877	0.921	0.927
5	9.930	$\mu_r + 0.5$	0.554	0.850	0.835	0.866	0.920	0.931
			Incongruence					
6	12.43	$\mu_r - 2.5$*	0.047	0.100	0.097	0.192	0.394	0.813
7	7.43	$\mu_r + 2.5$*	0.05	0.102	0.096	0.181	0.387	0.817
8	12.43	$\mu_r + 0.5$	0.564	0.619	0.588	0.426	0.255	0.136
9	12.43	μ_r	0.627	0.679	0.638	0.533	0.427	0.387
10	7.43	μ_r	0.629	0.678	0.642	0.537	0.425	0.374
11	7.43	$\mu_r - 0.5$	0.570	0.612	0.597	0.438	0.254	0.131

* Type I error.

1 with binary outcome) and omalizumab (trial 2 with continuous outcome) described previously. For trial 1, the hypotheses of non-inferiority test are: $H_0 : \mu_t/\mu_r \leq 0.8$ *vs.* $H_1 : \mu_t/\mu_r > 0.8$. We considered three values of reference means μ_r, which represent exact congruence (0.413) and incongruence in both directions (0.516 and 0.310). We considered three values of μ_t for the test arms: $\mu_t = 0.8\mu_r$ (inferiority) and $\mu_t = 0.95\mu_r$ or μ_r (noninferiority). For trial 2, the hypotheses of non-inferiority test are: $H_0 : \mu_t - \mu_r \leq -2.5$ *vs.* $H_1 : \mu_t - \mu_r > -2.5$. We considered three values of reference means μ_r, which represent exact congruence (9.930) and incongruence in both directions (12.43 and 7.43), and three values of μ_t: $\mu_t = \mu_r - 2.5$ (inferiority) and $\mu_t = \mu_r - 0.5$ or μ_r (noninferiority). The total sample size is 840 for trial 1 and 270 for trial 2. Patients are 1:2 randomized into reference and test arms.

Table 5 displays results of the noninferiority test for both binary and continuous endpoints. Scenarios 1 and 7 are used to calibrate the six designs such that they have type I errors of 0.05 when D is congruent to $\boldsymbol{D_0}$ and $\mu_t = 0.8\mu_r$. In general, the results are similar to those reported above for the equivalence test. When D and $\boldsymbol{D_0}$ are congruent, EMAP1 and EMAP2 provide substantial power gain over the NB method; when D and $\boldsymbol{D_0}$ are incongruent, EMAP methods offer better type I error control than MAP, rMAP50, and rMAP90. For example, in scenario 2, D and $\boldsymbol{D_0}$ are congruent and μ_t is noninferior to μ_r. EMAP1 and EMAP2 generate powers of 80.0% and 80.5%, whereas NB has a power of 61.5%. Similar results are observed in scenario 3. Scenarios 4–6 consider that D and $\boldsymbol{D_0}$ are incongruent. In scenario 4, μ_t is inferior to μ_r. EMAP1 and EMAP2 both control type I errors at 8.2%, whereas type I errors of rMAP50, rMAP90, and MAP are inflated to 16.0%, 32.0%, and 54.5%, respectively. In scenarios 5 and 6, μ_t is noninferior to μ_r. EMAP1 generates higher powers than rMAP50, rMAP90, and MAP. For example, in scenario 5, EMAP1 and EMAP2 have powers of 70.3% and 70.8%, whereas the powers of rMAP50, rMAP90, and MAP are 50.8%, 33.0%, and 21.1%.

For continuous endpoints, similar conclusions could be drawn. When D and $\boldsymbol{D_0}$ are congruent and μ_t is noninferior to μ_r (i.e., scenarios 8–9), EMAP1 and EMAP2 designs provide substantial power gain compared to NB. For example, in scenario 8, powers of EMAP1 and EMAP2 are 75.7% and 73.6%, respectively, about 22% higher than NB. When D is not congruent to $\boldsymbol{D_0}$, EMAP1 and EMAP2 control type I errors better than rMAP50, rMAP90, and MAP. For example, in scenario 10, type I errors of EMAP1 and EMAP2 are 10.2% and 9.6%, respectively, whereas the type I errors of rMAP50, rMAP90, and MAP are inflated as high as 20.1%, 42.8%, and 79.5%. Scenarios 11–12 show that when D is not congruent to $\boldsymbol{D_0}$ and μ_t is noninferior to μ_r, EMAP1 and EMAP2 generate higher power than rMAP50, rMAP90, and MAP. For example, in scenario 11, EMAP1 and EMAP2 generate powers of 68.2% and 67.8%, respectively, whereas the powers of rMAP50, rMAP90, and MAP are 52.9%, 37.5%, and 24.0%.

4. Conclusion

We have proposed an EMAP prior approach to dynamically incorporate multiple historical datasets in biosimilar and noninferiority trials. Our method is built upon the MAP prior, which extracts the prior information from historical datasets through meta-analysis. The

Table 5
Simulation results of non-inferiority test for binary and continuous endpoints.

			Probability of claiming biosimilarity					
Scenario	μ_r	μ_t	NB	EMAP1	EMAP2	rMAP50	rMAP90	MAP
Congruence				**Binary endpoints**				
1	0.413	$0.8\mu_r$*	0.050	0.050	0.050	0.051	0.050	0.050
2	0.413	$0.95\mu_r$	0.615	0.800	0.805	0.82	0.834	0.839
3	0.413	μ_r	0.826	0.942	0.946	0.952	0.969	0.967
Incongruence								
4	0.516	$0.8\mu_r$*	0.041	0.082	0.082	0.160	0.320	0.545
5	0.310	μ_r	0.651	0.703	0.708	0.508	0.330	0.211
6	0.310	$0.95\mu_r$	0.457	0.521	0.524	0.346	0.181	0.071
Congruence				**Normal endpoints**				
7	9.930	$\mu_r - 2.5$*	0.050	0.051	0.050	0.051	0.050	0.050
8	9.930	$\mu_r - 0.5$	0.540	0.757	0.736	0.803	0.858	0.866
9	9.930	μ_r	0.702	0.880	0.871	0.918	0.958	0.963
Incongruence								
10	12.43	$\mu_r - 2.5$*	0.050	0.102	0.096	0.201	0.428	0.795
11	7.43	μ_r	0.696	0.682	0.678	0.529	0.375	0.240
12	7.43	$\mu_r - 0.5$	0.533	0.506	0.519	0.352	0.209	0.080

* Type I error.

contribution of the EMAP prior method is to enable dynamic information borrowing from the MAP prior according to the congruence between the trial data and historical data. The EMAP prior method is information-borrowing consistent, asymptotically controls type I error, and is easy to quantify using prior effective sample size. Simulation study shows that the EMAP prior outperforms MAP prior and robust MAP priors, with better type I error control and competitive or higher power.

To design a biosimilar trial using the proposed methodology, the sample size can be determined as follows. We first elicit from clinicians the expected mean difference between the historical data and the reference arm, e.g., 25% of the standard deviation of the historical data. Given the expected mean of the reference arm, we simulate the reference data and calculate the congruence measure S. The effective sample size of the resulting elastic prior is $m \times g(S)$, where m is the sample size of the historical data. That is, the information expected to be borrowed from the historical data is equivalent to $m \times g(S)$ patients. Therefore, the sample size for the reference arm and the test product arm can be determined using the standard sample size calculation method (e.g., given specific type I and II errors), and then reduce the sample size of the reference arm by $m \times g(S)$. The sample size determination can be refined using the group sequential approach by updating the value of the congruence measure between the historical data and the reference arm, and thus the effective sample size, based on the interim data. If the congruence is less (or more) than expected (i.e., the mean difference between the historical data and the reference arm is greater (or smaller) than the expected), we can increase (or decrease) the enrollment for the reference arm. This will be the topics of our future research.

We have focused on biosimilar trials. The EMAP prior method, however, is general and does not hinge on a specific type of tests or trials. For example, it is directly applicable to randomized superiority clinical trials with historical data available for the control. The only modification is to replace the biosimilar test criteria with a superiority test criteria, for example, to claim that the treatment is superior to the control if $Pr(\mu_t > \mu_r|data) > C$, where μ_r now refers to the mean of the control. The EMAP prior method can also be easily modified to accommodate the time-to-event endpoint. We first employ any existing survival data meta-analysis method to obtain the MAP prior, and then use the logrank test statistic as the congruence measure S to construct the elastic function and EMAP prior. Lastly, the EMAP prior method can be combined with the group sequential method to design clinical trials. At each interim, we apply the EMAP and compare the treatment arm and the reference (or control) arm. Based on the strength of the evidence, appropriate interim decisions (e.g., futility or superiority stopping) can be made. One important advantage of the EMAP method is that it determines the information borrowing based on the congruence between historical data and trial interim data. When the trial proceeds, the degree of information borrowing will adaptively adjusted and updated based on the interim data. In contrast, other methods, e.g., power prior, MAP prior or robust MAP prior, will not make such adaptation as these priors are fixed.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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